

Colon adenocarcinoma

2nd leading cause of cancer death · ~106,590 new cases (2024) · 5-yr survival — stage I **90%**, II **70–85%**, III **50–73%**, IV **12–14%**

Stanford Oncology Handbook 2025, Colon Cancer (Johnson T, Miller D) · NCCN · items marked † are outside the handbook chapter

WORKUP

- Colonoscopy with biopsy + pathology review
- MMR / MSI testing in ALL newly diagnosed** colon cancer
- KRAS, NRAS, BRAF** mutation + **HER2** amplification by NGS panel
- CT C/A/P · MRI liver if potentially resectable liver mets · PET/CT in selected resectable M1
- Labs: CBC, CMP, **CEA**
- Genetics referral: all diagnosed **<50 y**, or somatic finding with germline implications
- BRAF V600E + MLH1 loss largely excludes Lynch — but ~1% of Lynch is BRAF V600E

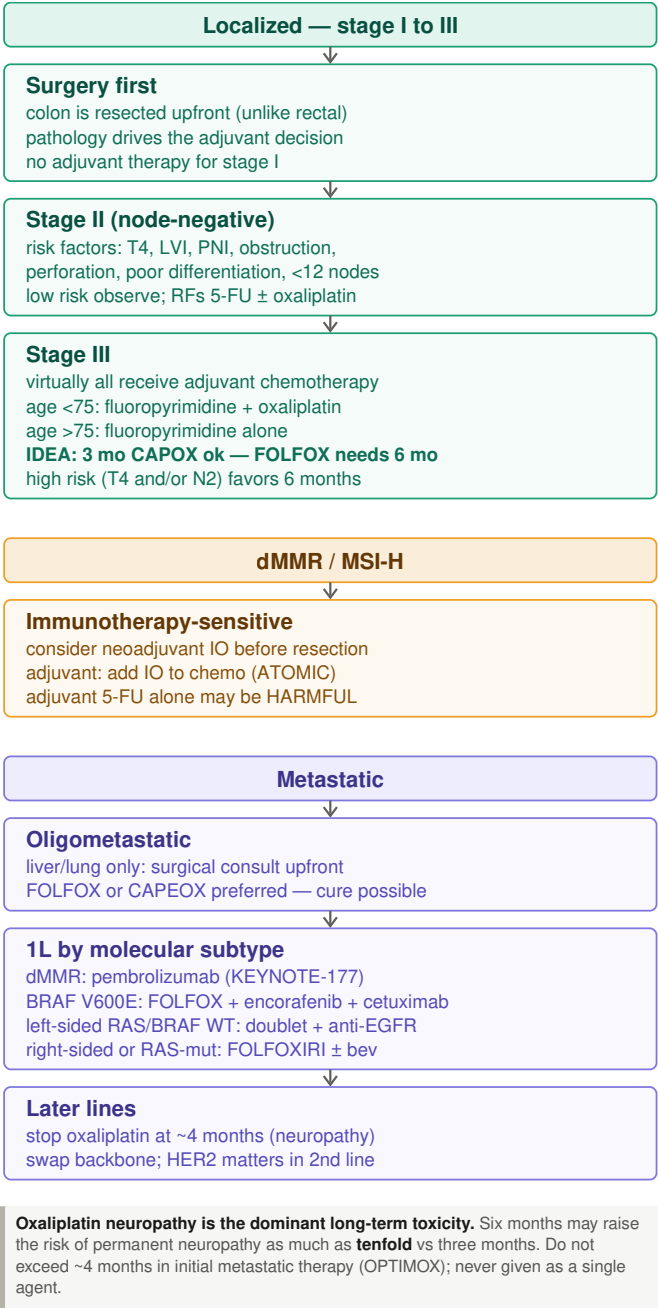
STAGING — AJCC 8TH

T1 / T2 / T3	submucosa / through muscularis propria / into pericorectal tissue
T4a / T4b	through visceral peritoneum / invades adjacent organ
N1 / N2	1–3 nodes (N1c = deposits only) / ≥4 nodes
M1a / b / c	1 site / ≥2 sites / peritoneal
Stage III	any node-positive, M0

KEY TRIALS

IDEA	3 vs 6 mo adjuvant — failed non-inferiority overall (HR 1.07, margin 1.12), but 3 mo CAPOX non-inferior while FOLFOX favored 6 mo (Grothey, NEJM 2018)
ATOMIC	adding IO to adjuvant chemo in dMMR (ASCO 2025)
KEYNOTE-177	1L pembrolizumab in dMMR: PFS 16.5 vs 8.2 mo
BREAKWATER	1L BRAF V600E: FOLFOX + encorafenib + cetuximab
BEACON	BRAF V600E later line (Kopetz, NEJM 2019)

TREATMENT PATHWAY



REGIMENS & DOSING

Regimen	Dosing	Cycle
FOLFOX	oxaliplatin 85 d1 · 5-FU 1200–1500/day CIV d1–2	q14d
CAPEOX / XELOX	oxaliplatin 130 d1 · capecitabine 1000 BID d1–14	q21d
FOLFIRI	irinotecan 180 d1 · 5-FU 1200–1400/day CIV d1–2	q14d
FOLFOXIRI	oxaliplatin 85 · irinotecan 165 · 5-FU 1200–1400/day CIV d1–2	q14d
Bevacizumab	7.5 mg/kg (q3wk) or 5 mg/kg (q2wk)	q21d / q14d
Cetuximab / panitumumab	cetuximab 400 load then 250 weekly (or 500 q2wk) · panitumumab 6 mg/kg	wkly / q2wk
Regorafenib	ramp 80→120→160 mg weekly in C1; then 160 mg d1–21	q28d
Trifluridine/tipiracil	35 mg/m ² PO BID d1–5 and d8–12	q28d
Fruquintinib	5 mg PO daily × 21 days	q28d

All doses mg/m² unless stated. **Institutional preference: no 5-FU bolus** (higher toxicity) — which also removes the need for leucovorin as a stabilizer.

BIOMARKER → THERAPY

Alteration	Therapy
dMMR / MSI-H	Pembrolizumab 200 mg q3wk or 400 mg q6wk — 1L metastatic; also neoadjuvant/adjuvant
BRAF V600E, 1L	FOLFOX + encorafenib 300 mg daily + cetuximab
BRAF V600E, later	Encorafenib + cetuximab; triplet adds binimetinib
RAS WT + left-sided	Cetuximab or panitumumab — no benefit if right-sided or RAS-mutant
HER2 amplified	Trastuzumab + pertuzumab; tucatinib 300 BID + trastuzumab; T-DXd
NTRK fusion	Larotrectinib 100 mg BID · entrectinib 600 mg daily

SURVEILLANCE

- H&P + **CEA** q3–6 mo × 2 yr, then q6 mo — 5 years total
- CT C/A/P: stage II–III q6–12 mo × 5 yr; stage IV q3–6 mo × 2 yr then q6–12 mo
- Colonoscopy at **1 year** post-op; within 3–6 mo if no complete pre-op scope
- If adenoma found repeat in 1 yr; otherwise 3 yr, then every 5 yr

PEARLS & CAUTIONS

- Irinotecan** with great caution in the elderly, biliary obstruction, or hyperbilirubinemia
- Regorafenib** is very toxic — many do better at lower doses; **bevacizumab** benefit is modest
- ctDNA timing conflict**: results take 4–6 wk but adjuvant works best within 6 wk of surgery
- Metastatic **rectal** regimens are identical to metastatic CRC